

- Q.29 Explain in brief the Electromagnetic flow meter. (CO6)
- Q.30 With the help of diagram, explain thermocouples. (CO7)
- Q.31 Show how Pyrometers work? (CO7)
- Q.32 Explain any one method of measurement of level. (CO8)
- Q.33 Show how torque is measured? (CO4)
- Q.34 Show how thermistors are used for measurement of temperature? (CO7)
- Q.35 What is bellow? How it is made? (CO5)

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 a) Explain in brief the Hall effect transducers. (6) (CO3)
- b) Define Linearity & Sensitive of transducers. (4) (CO2)
- Q.37 With the help of neat diagram, explain bimetallic thermometer. (CO7)
- Q.38 Draw and explain the principle and working of LVDT. (CO3)

No. of Printed Pages : 4

181041/121043/062444

Roll No.

/030954-A/105955/031062

4th Sem /Eltx, Med Eltx, Mecatronics, Power Eltx
Subject:- Instrumentation / Instrument, Process Control

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 Thermistor is _____ Transducer. (CO7)
- a) Inductive b) Resistive
- c) Capacitive d) Piezo-eletric
- Q.2 Unit of Force is _____. (CO4)
- a) Newton b) Newton-m
- c) Newton/m d) Newton-m²
- Q.3 Thermocouple works on the principle of ____ (CO7)
- a) Variable resistance b) Seebach effect
- c) Doppler effect d) Faraday's law
- Q.4 Dynamometer is used to measure ____ (CO4)
- a) Force b) Strain
- c) Torque d) Temperature
- Q.5 Humidity is measured by ____ (CO8)
- a) Potentiometer b) Bourdon tube
- c) Hygrometer d) Thermometer

(900)

(4) 181041/121043/062444
 /030954-A/105955/031062

(1) 181041/121043/062444
 /030954-A/105955/031062

- Q.6 Electromagnetic flow meter depends on density of liquid (CO6)
 a) True b) False
- Q.7 Flow is expressed in _____ (CO8)
 a) Liters-minute b) Litres-minute²
 c) Liters/minute d) None
- Q.8 Strain Gauge is _____ Transducer. (CO3)
 a) Active b) Passive
- Q.9 pH of acids is _____ (CO8)
 a) ZERO b) 7
 c) >7 d) <7
- Q.10 _____ material is used for resistance thermometers? (CO7)
 a) Iron b) Copper
 c) Platinum d) Aluminum

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Define Signal conditioning. (CO1)
- Q.12 Name any one piezoelectric material. (CO2)
- Q.13 What are units of strain. (CO3)
- Q.14 Define Humidity. (CO8)
- Q.15 Define Seeback effect. (CO7)

- Q.16 What is use of Load cells? (CO4)
- Q.17 Define Torque. (CO4)
- Q.18 Name the material used for making bourdon tubes. (CO5)
- Q.19 Expand LCD. (CO1)
- Q.20 Define Gauge factor. (CO3)

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Write different stages of a measurement system. (CO1)
- Q.22 Write in detail the working principle of Capacitive transducers. (CO2)
- Q.23 With examples, Differentiate between Active & Passive transducers. (CO1)
- Q.24 Explain basic principle of piezoelectric transducers. (CO2)
- Q.25 Show how displacement is measurement by potentiometers. (CO3)
- Q.26 Write any one method for measurement of humidity. (CO8)
- Q.27 Explain in brief the working of pressure cells. (CO5)
- Q.28 Explain in brief about measurement of speed. (CO8)